Hotel Booking

1. Home page

Description: The home page is the main interface of the hotel booking website. This page includes an introduction section, filters for city and booking date, login and registration functions, a complete list of rooms, as well as navigation options**.**

Actors: Guest, Customer.

Preconditions: The user accesses the website.

Main Flow:

1. The user opens the website.
2. The system displays the home page.
3. The system shows the hotel introduction, filters for searching rooms by city and booking dates, login and registration functions, and a complete list of available rooms.
4. The system displays navigation options such as Home, Rooms, My Bookings, Login, and Register.
5. The user selects a function to continue, such as searching for rooms, viewing room details, logging in, or registering a new account.

Postconditions: The home page is displayed, and the user can navigate to other functions.

2. View home

Description: This function allows users to view the home page and filter rooms by room code and check-in date. It also provides sign-in and sign-up functions.

Actors: Guest, Customer.

Preconditions: The website is accessible.

Main Flow:

1. The user accesses the website.
2. The system displays the home page.
3. The system displays the room filtering options, including room code and check-in date.
4. he system displays the sign-in and sign-up functions..
5. The user selects a function to continue.

Alternative Flow:

- If room information cannot be loaded, the system displays an error message.

- If no promotions are available, the system displays the home page without promotional content.

Postconditions: The user can view the available information and choose the next action.

3. Booking page

Description: This function allows customers to view room details, book a room, apply a voucher, and enter or update customer information.

Actors: Customer.

Preconditions:

- The customer has selected a room.
- The selected room is available for the requested dates.

Input: Room information, check-in date, check-out date, number of guests, customer information, and voucher code (optional).

Main Flow:

1. The customer selects a room.
2. The system displays the room details and booking page.
3. The customer enters the check-in date, check-out date, and number of guests.
4. The customer enters or updates their personal information.
5. The customer enters a voucher code if available.
6. The system validates the booking information and voucher code.
7. The system calculates the total booking cost.
8. The customer reviews and confirms the booking information.
9. The system saves the booking information and proceeds to the payment page.

Alternative Flow:

- If required information is missing or invalid, the system asks the customer to correct it.
- If the room is unavailable, the system asks the customer to select another room or date.
- If the voucher code is invalid, the system displays an error message.

Postconditions: The booking information is recorded, and the customer can proceed to payment.

4. Payment page

Description: This function allows customers to view their booking information and make online payments for their hotel reservations.

Actors: Customer, Payment Page.

Preconditions:

- The customer has confirmed the booking details.
- The system has calculated the total amount.

Input: Booking information, payment amount, and payment method.

Main Flow:

1. The system displays the booking summary and total amount.
2. The customer reviews the payment information.
3. The customer selects an available payment method.
4. The customer confirms the payment.
5. The system sends the payment request to the payment gateway.
6. The payment gateway processes the transaction.
7. The system receives the payment result.
8. If the payment is successful, the system updates the booking status and displays the payment successful page.

Alternative Flow:

- If the payment fails, the system displays an error message.
- If the customer cancels the payment, the system returns to the appropriate booking or payment screen.
- If the payment gateway is unavailable, the system informs the customer that the payment cannot be processed at that time.

Postconditions: The payment result is recorded. If successful, the booking is marked as paid or confirmed according to the system's booking rules.

5. Payment successful

Description: This function allows customers to view payment information, receive an invoice, and confirm successful payment for their hotel booking.

Actors: Customer, Payment page.

Preconditions:

- The customer has completed the online payment process.
- The payment gateway has returned the transaction result.

Input data: Booking information, payment transaction details, and invoice information.

Main Flow:

1. The system receives the successful payment result.
2. The system updates the payment status.
3. The system confirms the booking according to the booking rules.
4. The system displays a payment success message.
5. The system displays the booking ID, room details, check-in date, check-out date, and total amount paid.
6. The customer can choose to view their bookings or return to the home page.

Alternative Flow:

- If the payment result cannot be verified, the system displays a pending status and asks the customer to check again.
- If the confirmation page cannot be loaded, the system preserves the recorded payment result so the customer can check the booking later.

Postconditions: The payment status is recorded, and the invoice is generated if the payment is successful.

6. Login

Description: This function allows registered customers to log in to their accounts and reset their passwords if they forget them.

Actors: Customer.

Preconditions: The customer has registered an account.

Input: Email address and password. For password reset, the customer provides the registered email address.

Main Flow:

1. The customer opens the login page.
2. The system displays the login form.
3. The customer enters their email address and password.
4. The customer submits the login form.
5. The system validates the entered information.
6. The system checks the credentials against the account records.
7. If the credentials are correct, the system creates an authenticated session.
8. The system redirects the customer to the home page or the previously requested page.

Alternative Flow:

- If the email address or password is incorrect, the system displays an error message.
- If the customer forgets their password, they select the “Reset password” function.

- The system asks the customer to enter their registered email address and sends instructions to reset the password
- If the account is unavailable or restricted, the system displays an appropriate message.

Postconditions: The customer can log in successfully or follow the password reset instructions to regain account access.

7. Register

Description: This use case allows new users to create an account on the website to manage their hotel bookings.

Actors: Guest.

Preconditions: The guest accesses the website and does not already have an account using the same email address.

Input: Full name, email address, password, and password confirmation.

Main Flow:

1. The guest opens the registration page.
2. The system displays the registration form.
3. The guest enters the required personal information.
4. The guest confirms the password and submits the form.
5. The system validates the submitted information.
6. The system checks whether the email address is already registered.
7. The system creates a new customer account if all information is valid.
8. The system displays a registration success message.
9. The guest can proceed to the login page.

Alternative Flow:

- If required information is missing, the system asks the guest to complete the form.
- If the email address is already registered, the system displays an error message.
- If the passwords do not match or fail the password requirements, the system asks the guest to enter a valid password.

Postconditions: A new customer account is created successfully, and the user can log in to the website.